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EXTENDED ABSTRACT

How Well Do Machine-Learning Weather Models Predict Atmospheric Rivers?

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1. Introduction and objectives

Atmospheric Rivers (ARs) are elongated corridors of intense water vapour transport that account for a large fraction of poleward moisture flux and are responsible for many high-impact precipitation events, particularly along the western margins of continents. Accurate prediction of AR position, orientation, timing, and intensity is therefore central to water-resource management, flood risk mitigation, and sub-seasonal hazard preparedness. Despite steady improvements in numerical weather prediction (NWP), AR forecasts remain challenging due to their filamentary structure, sharp gradients, and sensitivity to errors in large-scale flow and moisture transport.

Machine learning weather prediction (MLWP) has demonstrated skill comparable to or exceeding NWP for many large-scale meteorological variables. Models such as GraphCast, NeuralGCM, and Pangu-Weather have shown impressive performance in medium-range forecasting benchmarks. However, their ability to represent coherent, mesoscale, extreme phenomena such as ARs, has not been systematically evaluated. Since ARs are defined by the coupled transport of moisture by the horizontal wind field, they provide a stringent test of whether machine learning models capture physically consistent inter-variable relationships between wind components and humidity. Existing evaluations largely rely on grid-point metrics (e.g. RMSE, ACC) that are poorly suited to diagnosing errors in spatially coherent objects, and therefore provide limited insight into the physical failure modes of MLWP models.

2. Methods and data

We present a unified verification framework to assess AR prediction skill, and apply it to several state-of-the-art MLWP models. The analysis focuses on the Northern Pacific basin (10°N–60°N, 180°W–110°W) during 2020, a region characterised by frequent landfalling AR events [Lam et al.(2023)].

We evaluate forecasts from NeuralGCM [Kochkov et al.(2024)], GraphCast [Lam et al.(2023)], Pangu-Weather [Bi et al.(2023)], and Keisler [Keisler(2022)] using ERA5 reanalysis as the reference dataset. All forecasts are obtained from the WeatherBench2 [Rasp et al.(2024)] archive and are regridded to a common horizontal resolution (0.7°) to enable consistent comparisons. ARs are diagnosed using Integrated Vapor Transport (IVT), computed from vertically integrated moisture flux between 1000 and 300 hPa using wind components and specific humidity. AR detection follows a threshold-based approach, with objects identified via connected-component labelling applied to IVT fields exceeding the 85th-percentile threshold at each time step. The detected regions are further filtered using

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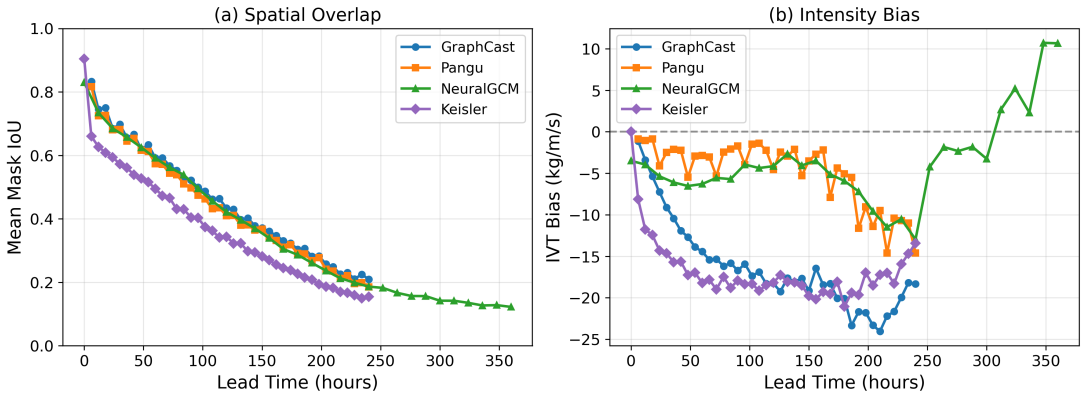

Figure 1. *AR verification metrics vs lead time: (a) spatial overlap (IoU) between forecast and ERA5-detected ARs, and (b) IVT intensity bias (forecast minus reference).*

geometric criteria, requiring a minimum filament length of 2000 km and an aspect ratio of at least 2.0 [Ralph et al.(2019)]. Object-based verification matches forecast and reference AR objects using Intersection-over-Union (IoU), with a minimum matching threshold of 0.1, IVT intensity bias within matched AR objects, and geometric errors, including length bias and orientation error.

3. Preliminary results

Figure 1 shows the spatial overlap and intensity bias between forecast and ERA5-detected AR as a function of lead time. All models exhibit a monotonic decline in spatial skill, with the IoU decreasing from 0.68–0.83 at 24 hours to 0.15–0.21 at 240 hours, corresponding to a 70–75% reduction over the forecast horizon of 10 days. Models retain skillful AR detection out to approximately 5 days (IoU > 0.40), with degraded but non-negligible skill extending to 10 days. In particular, spatial coherence degrades more slowly than the AR intensity, indicating that ML models better preserve the AR geometry than the magnitude of moisture transport.

All models exhibit a systematic negative IVT bias that increases with lead time (barring NeuralGCM beyond 300 hours), ranging from 4 to 14 $\mathrm{kg\,m^{-1}s^{-1}}$ at 24 houtd and 13 to 18 $\mathrm{kg\,m^{-1}s^{-1}}$ at 240 hours. This behaviour is consistent with smoothing of joint wind–moisture extremes and regression toward climatological means, suggesting limitations in capturing physically consistent inter-variable coupling between wind and humidity. Keisler shows the largest bias, likely due to missing lower-tropospheric moisture contributions below 500 hPa.

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